

BACHELORS ISLAND, MB, Canada

WMO: 711430

Lat: 51.750N

Lon: 99.900W

Elev: 840

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 94-10

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-22.1	-32.8	0.9	-26.9	-27.8	1.2	-22.0	31.9	17.8	27.8	7.6	6.4	320	0.749

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	79.9	71.0	77.5	69.9	75.3	68.6	72.9	78.0	71.0	76.1	69.2	74.3	11.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.1	118.2	77.4	69.1	110.3	75.3	67.2	103.1	73.4	37.2	78.1	35.5	76.0	33.9	74.6	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	29.7	26.3	23.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)				N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	-0.9	7.3	19.5	33.6	48.3	61.4	67.8	66.3	56.5	43.0	25.0	9.6	(6)
(7)		DBStd	25.31	13.61	13.81	14.24	9.03	8.76	6.54	4.76	5.84	7.46	6.05	9.81	13.89	(7)
(8)		HDD50	6611	1579	1194	945	494	140	4	0	0	25	227	749	1254	(8)
(9)		HDD65	10626	2044	1614	1411	943	520	141	27	55	270	683	1199	1719	(9)
(10)		CDD50	1718	0	0	0	1	87	347	551	504	219	9	0	0	(10)
(11)		CDD65	256	0	0	0	0	2	34	112	94	14	0	0	0	(11)
(12)		CDH74	770	0	0	0	0	11	88	344	305	22	0	0	0	(12)
(13)		CDH80	79	0	0	0	0	1	11	28	38	1	0	0	0	(13)
(14)	Wind	WSAvg	11.4	9.7	10.1	10.6	10.8	12.0	11.8	11.3	12.1	13.1	12.9	12.1	10.7	(14)
(15)	Precipitation	PrecAvg	20.5	1.1	0.6	1.2	1.1	2.0	3.3	3.0	2.4	2.0	1.4	1.0	1.2	(15)
(16)		PrecMax	24.0	3.0	1.2	2.4	2.0	4.5	5.0	5.2	4.3	4.0	2.4	3.6	2.2	(16)
(17)		PrecMin	17.2	0.2	0.0	0.4	0.2	0.6	1.2	1.2	1.1	0.4	0.4	0.2	0.4	(17)
(18)		PrecStd	1.6	0.6	0.3	0.7	0.6	1.0	1.2	1.1	0.9	1.1	0.6	0.8	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	35.8	48.1	58.9	74.3	81.0	83.2	83.7	77.3	59.8	46.4	40.3	(19)
(20)			MCWB	36.5	34.0	40.4	47.7	59.8	69.9	73.2	72.7	66.4	53.4	40.6	36.2	(20)
(21)		2%	DB	30.6	33.9	41.4	53.0	67.5	76.2	79.9	80.0	72.9	55.5	42.1	35.6	(21)
(22)			MCWB	29.4	32.7	37.3	45.7	59.0	68.8	71.3	71.6	66.0	49.2	37.7	33.4	(22)
(23)		5%	DB	24.0	32.2	38.9	48.0	63.5	73.3	77.8	77.2	70.3	53.0	39.1	30.6	(23)
(24)			MCWB	23.0	31.2	36.1	42.7	56.9	66.6	70.2	70.3	64.4	48.3	36.2	29.0	(24)
(25)		10%	DB	19.3	27.9	36.5	44.0	60.4	70.7	75.5	74.9	67.3	51.0	37.1	27.5	(25)
(26)			MCWB	18.6	27.0	34.4	39.9	54.5	64.9	68.9	68.8	61.8	47.4	34.9	26.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	34.2	40.9	49.6	62.8	73.4	75.6	75.2	68.3	54.1	41.0	36.3	(27)
(28)			MCD B	39.1	34.9	46.3	56.1	70.6	77.2	80.6	81.6	73.9	58.7	45.3	39.3	(28)
(29)		2%	WB	29.6	32.8	37.9	46.4	60.2	70.0	72.9	73.2	66.9	51.1	38.4	33.2	(29)
(30)			MCD B	30.6	33.8	40.9	51.5	65.8	74.4	78.1	78.5	72.3	54.0	41.5	35.7	(30)
(31)		5%	WB	23.2	30.9	36.0	43.2	57.5	67.6	71.1	71.1	64.9	49.6	36.6	29.2	(31)
(32)			MCD B	24.1	32.0	38.3	48.4	63.2	72.3	76.5	75.9	70.0	52.3	39.1	30.7	(32)
(33)		10%	WB	18.7	27.0	34.4	39.9	54.7	65.5	69.4	69.2	62.3	48.0	34.9	26.4	(33)
(34)			MCD B	19.3	27.9	36.2	43.7	59.8	70.0	74.7	74.2	66.9	50.9	37.0	27.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.1	17.7	17.6	14.9	11.8	10.7	11.4	10.7	10.0	8.3	9.7	12.8	(35)
(36)			MCDBR	16.4	14.6	18.0	19.1	16.5	15.1	14.4	13.7	14.1	12.2	11.3	16.2	(36)
(37)			MCWBR	15.6	13.8	14.6	12.6	9.8	10.0	9.7	8.7	9.2	8.5	8.9	15.0	(37)
(38)		5% WB	MCD BR	15.6	14.2	16.9	18.8	13.6	13.2	13.5	12.9	12.8	11.6	11.2	16.1	(38)
(39)			MCWBR	14.9	13.5	14.0	12.6	8.8	9.5	9.6	9.0	9.0	8.5	9.2	14.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.266	0.273	0.290	0.352	0.361	0.382	0.385	0.375	0.342	0.296	0.265	0.249	(40)	
(41)		taud	2.237	2.261	2.364	2.354	2.393	2.361	2.385	2.405	2.483	2.570	2.433	2.282	(41)	
(42)		Ebn at Noon	236	271	290	281	283	277	274	271	268	265	241	225	(42)	
(43)		Edh at Noon	21	27	30	35	36	37	36	33	27	21	17	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	370	739	1194	1591	1759	1815	1909	1566	1124	638	333	263	(44)	
(45)		RadStd	31	48	82	117	156	132	87	122	105	82	45	26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	-0.81	-1.01	N/A	N/A	N/A	N/A

Nomenclature: See separate page